Sagar Gupta

ProServe (Cloud Consultant) - DevOps/MLOps | Amazon Web Services (AWS) | Hyderabad, India

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Summary

Senior DevOps Consultant at AWS Professional Services with two years shipping CI/CD, IaC, and multi-account governance for enterprise financial-services and technology customers (RWS, State Street, DTCC). Deep Terraform (modules, tftest, PyTest) plus GitHub Actions, CodePipeline, CodeDeploy, and blue-green delivery on ECS. Built Terraform with automated tests for about 10 AWS services at State Street, and now automate RWS's AWS foundation in Terraform Cloud, cutting per-account setup from 3+ days to under 4 hours. 11 upstream PRs merged across axios, Airflow, lucide, Prefect, Feast, awslabs/mcp, multer, and others. 5x AWS certified. LeetCode Guardian – 2166 peak, 1300+ problems, 107 contests.

Experience

Amazon Web Services (AWS) Aug 2024 – Present
ProServe (Cloud Consultant) - DevOps/MLOps Hyderabad, India

- **Lead DevOps Consultant – RWS (Feb 2026 – Present):** Replaced a legacy account-vending pipeline with Terraform Cloud automation (OIDC) for a data-centre exit and a second landing zone, cutting per-account setup from 3+ days to under 4 hours (~90% reduction). Codified Organizations OUs with Control Tower baselines, SCPs, tag policies, and least-privilege account baselines with IAM Identity Center, AWS Backup, and SSM Patch Manager in every workload account: 4,000+ resources across 20+ Terraform Cloud workspaces, none failing. Applied every change in a sandbox mirror first through the same least-privilege role. Built self-service network onboarding on two Transit Gateway hubs: route table within 15 minutes, firewall access by tag in about a minute. Used Kiro and Claude Code for specs and design documentation.
- **DevOps Consultant – State Street (Oct 2024 – Aug 2025):** Built Terraform for about 10 AWS services with automated tests (tftest + PyTest) on the regulated landing zone team, part of the path that opened 38+ services to application teams. Designed Amazon Detective enablement via an AFT customization, with organization-wide auto-enable in Terraform. Worked on the security solution behind 1,000+ preventative CCMv4.0 controls (team result). Project closed with a **10/10 CSAT**.
- **DevOps Consultant – DTCC (Aug 2025 – Nov 2025):** Modernized legacy Terraform into cloud-native IaC patterns on high-sensitivity financial workloads – refactored monoliths into reusable modules, added remote state with S3 native locking, and hardened pipelines with policy-as-code checks. Used Amazon Q to accelerate code analysis and migration assessments across multiple workstreams.
- **MLOps Pipeline (SME Program, Jun – Aug 2025):** Built end-to-end SageMaker pipeline for medical-image classification with blue-green endpoint deployment, auto-scaling, and drift detection – provisioned via Terraform + CodePipeline with automated rollback on canary failure. Generalized and published by AWS as an official open-source sample (aws-samples, MIT-0).
- **Internal Impact:** Published a second AWS sample, Terraform for AWS Organizations governance on Control Tower (aws-samples, MIT-0), covering all six policy types, Control Tower controls, and delegated admin for 15 services. Published APG pattern *Secure PR Previews and Approval-Gated Releases for Amplify Gen 2 with GitHub* (plus 2 more peer-reviewed in the publication pipeline). Delivered 2 org-wide tech talks on Terraform module testing and GitOps pipeline gating. 5x TFC Ambassador (AI/ML, NGDE Graduated, CloudOps – AWS DevOps Agent, Agentic AI, Serverless). MLOps SME Program Graduate. Associate Speaker Certified.

Amazon Web Services (AWS) Mar 2024 – Jul 2024
Associate Professional Services – DevOps (Internship) Hyderabad, India

- Migrated on-premises application to AWS ECS with blue-green deployment via Terraform, CodePipeline, and CodeDeploy – enabling zero-downtime releases. Instrumented X-Ray and CloudWatch distributed tracing for cross-service latency visibility.

Ikarus-3D May 2023 – Jul 2023
Software Developer Associate (Internship) Mohali, India

- Containerized production application on AWS ECS/ECR with GitHub Actions CI/CD, and deployed SonarQube on EC2 + RDS to embed continuous code-quality analysis into the delivery pipeline.

Industry Certifications (5)

AWS: Solutions Architect – Associate | Developer – Associate | Machine Learning Engineer – Associate
AWS: AI Practitioner | Cloud Practitioner

Education

National Institute of Technology Warangal, Telangana Sep 2021 – Jul 2024
Master of Computer Applications (MCA) | NIMCET AIR 208 CGPA: 8.38

Sanskar College of Professional Studies (DAVV University), Indore May 2018 – Aug 2021
Bachelor of Computer Applications (BCA) CGPA: 7.34

Projects

Blue-Green AWS Terraform Jul 2025
Terraform, AWS ECS on EC2, ALB, CodePipeline, HCL GitHub
◇ Production-grade IaC for zero-downtime blue-green deployments on ECS backed by an EC2 Auto Scaling Group, across 15 Terraform modules, with KMS encryption, ALB weighted traffic shifting, and least-privilege IAM.
◇ Full pipeline via CodePipeline, CodeBuild, and CodeDeploy for automated image builds, test gates, and progressive traffic cut-over between blue and green target groups with rollback on alarm.

DevOps-AWS-FARM Aug 2024
Terraform, Docker, GitHub Actions, FastAPI, React, MongoDB GitHub
◇ Full-stack FARM application with production DevOps pipeline – multi-stage Docker builds, GitHub Actions CI/CD, Terraform-provisioned infrastructure, and automated ECS deployments with health-check gating.

GitScope Mar 2026
JavaScript, Chrome Extension (Manifest V3), GitHub GraphQL, Cloudflare Workers GitHub | Chrome Web Store
◇ Chrome extension adding contribution insights to any GitHub profile – streaks, language breakdown, PR stats, and activity heatmap. OAuth proxy on Cloudflare Workers, 5-minute edge caching, zero runtime dependencies.

Ledger Sync Jan 2026
TypeScript, React, FastAPI, SQLite Live
◇ Personal-finance transaction reconciler with rules-based categorization, duplicate detection across statement imports, and a typed FastAPI backend served behind a static React UI.

Other Notable Projects: Claude Cost Optimizer (30–60% Claude spend cuts via model routing + prompt-cache tuning), Bedrock Multi-Model MCP (500+ models via one server), LeetCode Rating Predictor (121K+ contest records, MAE ~7.8), Instagram Likes Leaderboard.

Open Source Contributions

- **aws-ia/terraform-aws-vpc** – Root-caused the Cloud WAN attachment-replacement bug on `create_vpc = false`; fix co-authored into v4.8.0 (merged)
- **hashicorp/terraform-provider-aws** (10K☆) – Validate IoT CloudWatch metric substitution templates (#46867, open)
- **terraform-aws-modules/atlantis** – Pass custom egress rules through to the ECS service security group (#446, merged)
- **axios** (109K☆) – Synthesize AxiosError message from AggregateError on dual-stack connection failures (#11059, merged)
- **apache/airflow** (44K☆) – Add `template_fields` to SalesforceBulkOperator (#63109, merged)
- **PrefectHQ/prefect** (21K☆) – Document deployment steps as Python functions (#20956, merged)
- **feast-dev/feast** (6K☆) – Ship Claude Code agent skills for Feast (#6081, merged)
- **awslabs/agent-plugins** – Add `defusedxml` extension guard (#132, merged)
- **awslabs/mcp** – Fix Kendra documentation menu to match the source directory (#2607, merged)
- **expressjs/multer** – Stop files skipped by `fileFilter` from consuming `maxCount` (#1426, merged)
- **lucide** (24K☆) – Support kebab-case icon-node prop on the Vue Icon component (#4512, merged)

Technical Skills

IaC & Config: Terraform (modules, tftest, PyTest, remote state, S3 native locking), HCL, CloudFormation, Ansible, Packer
CI/CD & Delivery: GitHub Actions, AWS CodePipeline, CodeBuild, CodeDeploy, Blue-Green & Canary Deployments, GitOps, Policy-as-Code
AWS Platform: Organizations, SCPs, Control Tower, Transit Gateway, Direct Connect, VPN, ECS Fargate, Lambda, EventBridge, API Gateway, CloudFront, KMS, IAM, Config, Security Hub
Observability & SRE: CloudWatch (Logs, Metrics, Alarms), X-Ray, SonarQube, Drift Detection, Auto-Scaling, Incident Runbooks
Languages & Runtime: Python, Bash, TypeScript, SQL, Docker, Linux
Compliance & Governance: CCMv4, Least-Privilege IAM, Preventative Controls, Tag Policies, Cost & Usage Reporting

Achievements

- Published APG pattern *Secure PR Previews and Approval-Gated Releases for Amplify Gen 2 with GitHub* (2 more peer-reviewed in the publication pipeline). MLOps SME Program Graduate. Associate Speaker Certified.
- 5x TFC Ambassador – AI/ML, NGDE (Graduated), CloudOps (AWS DevOps Agent), Agentic AI, Serverless.
- LeetCode Guardian – 2166 peak rating, 1300+ problems solved, 107 contests. GfG: 500+ DSA problems. HackerRank: 6-Star Problem Solving, 5-Star C++. *LeetCode*
- Rank 1289 (Top 5%) – Google Kick Start Round E '22. *Certificate*
- 1st place – Coding Cauldron '23 (NIT Trichy) and QFiesta IGNOSI '22 (NIT Warangal).